CHAPTER 2

Quadratics: forms and choice of method

Key point · Chapter path. **Objectives.** Choose a form; solve an equation or an inequality; read an extremum; prove the formula for the roots.

Prerequisites. Special products, zero-product property, sign of a product, square roots.

Organisation. Practise fluency, investigate, formalise, then apply.

Diagnostic: six questions, 10 minutes

Answer without consulting the lesson. Mark the questions for which help is needed. This diagnostic is not graded.

1. Expand $(x-3)^2$.
2. Factorise x^2-16.
3. Solve $(x-2)(x+5)=0$.
4. Solve $x^2=9$ in $\mathbb{R}$.
5. Give the sign of $2x-6$.
6. Find the minimum of $(x-1)^2+4$.

After the diagnostic. Revisit the flagged topics with the teacher; redo a similar question, then explain the rule used.

Speaking and logic. Present two methods for factorising a quadratic. In the proof using the discriminant, name the three cases and explain why they are exhaustive.

Investigate before formalising

A rectangle has perimeter 24 units. Choose several positive widths that give a rectangle. For widths 2, 4, 6, 8 and 10, calculate the length and area. Which width seems to give the greatest area? State a conjecture, then seek a justification valid for every admissible width.

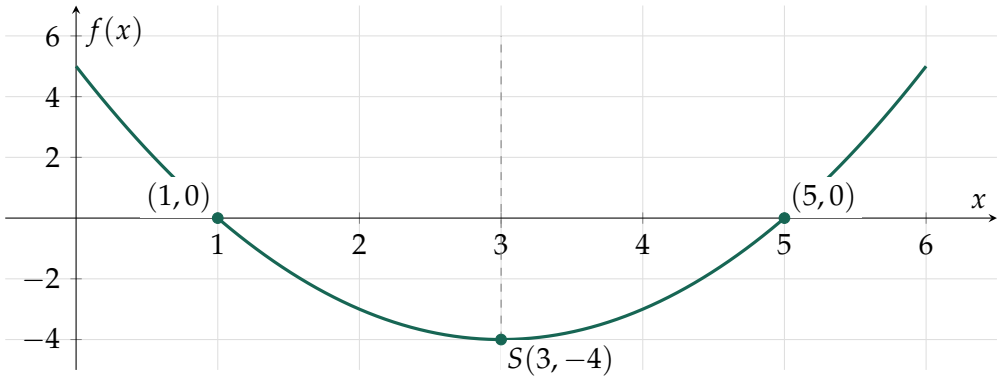
Organisation. Individual investigation 5 min, comparison in pairs 5 min, class discussion 5 min. Keep a record of one attempt, even an unfinished one. A conjecture must then be justified.

2.1 Three forms, three readings

Key point · Definition and results. A quadratic function is written $f(x) = ax^2 + bx + c$ with $a \neq 0$. The expanded form makes it easy to calculate $f(0)$; the vertex form $a(x-\alpha)^2 + \beta$ reveals the vertex and the extremum; a factorised form $a(x-r_1)(x-r_2)$ reveals the roots and the sign. The roots are the solutions of $f(x) = 0$.

Status and justification. Completing the square relies on $x^2 + 2ux = (x+u)^2 - u^2$. The existence of the vertex form is proved by expansion; its general computation is not an expected routine.

Example 1. $f(x) = x^2 - 6x + 5 = (x-3)^2 - 4 = (x-1)(x-5)$. The minimum is -4 at 3; the roots are 1 and 5; the y-intercept is 5.



Graph of $f(x) = (x-3)^2 - 4$. The two roots and the vertex are marked; the axis of symmetry is the line $x = 3$.

Watch out. A factorised expression is not obtained by putting each term separately in brackets.

Application. Give three forms of $g(x) = x^2 + 2x - 3$. Read off its roots and its minimum.

2.2 Discriminant and solving

Key point · Definition and results. For $ax^2 + bx + c = 0$ with $a \neq 0$, let $\Delta = b^2 - 4ac$. If $\Delta < 0$, there is no real root; if $\Delta = 0$, there is one root $-\dfrac{b}{2a}$; if $\Delta > 0$, there are two roots $\dfrac{-b - \sqrt{\Delta}}{2a}$ and $\dfrac{-b + \sqrt{\Delta}}{2a}$. To order the roots, take the sign of a into account.

Status and justification. Multiplying the equation by $4a \neq 0$ gives $(2ax + b)^2 = \Delta$. A square is positive or zero: the three cases follow. If $\Delta > 0$, $2ax + b = \pm\sqrt{\Delta}$. Splitting into cases avoids taking the square root of a negative number.

Example 2. $2x^2 + x - 3 = 0$: $\Delta = 1 + 24 = 25$; $x = \dfrac{-1 \pm 5}{4}$, that is 1 and $-\dfrac{3}{2}$. Check: $(x-1)(2x+3) = 2x^2 + x - 3$.

Watch out. The denominator is $2a$, not 2; the sign of b is kept with its brackets in b^2.

Application. Solve $3x^2 - 6x + 3 = 0$ and $x^2 + 2x + 5 = 0$.

2.3 Factorising without automatically using the discriminant

Key point · Definition and results. If two real roots exist (possibly equal, in which case they are counted twice), their sum is $-\frac{b}{a}$ and their product is $\frac{c}{a}$. An obvious root or a remarkable identity often gives a shorter factorisation. The quadratic polynomials vanishing at two distinct reals r and s are exactly $a(x-r)(x-s)$ with $a \neq 0$.

Status and justification. Expanding $a(x-r)(x-s) = ax^2 - a(r+s)x + ars$ and comparing coefficients gives the relations. Conversely, if r and s have this sum and this product, expanding recovers the polynomial.

Example 3. Let $P(x) = x^2 - 7x + 12$. The numbers 3 and 4 have sum 7 and product 12, hence $P(x) = (x-3)(x-4)$. A separate question: determine a function f with roots 3 and 4 and $f(0) = 24$. Write $f(x) = a(x-3)(x-4)$. Then $12a = 24$, so $a = 2$: f differs from P.

Watch out. The sum of the roots is $-\frac{b}{a}$; do not forget the minus sign. The roots do not determine a: an extra condition is needed, for example the value of f at one point.

Application. Determine the polynomial f of degree 2 such that $f(1) = f(4) = 0$ and $f(0) = -8$.

2.4 Sign, inequalities and optimisation

Key point · Definition and results. For two distinct roots $r < s$, the quadratic has the sign of a outside $[r, s]$ and the opposite sign between the roots. It is zero at the roots. With no real root, it keeps the sign of a. With a double root, it keeps the sign of a away from that root. Optimisation is done on the domain of the problem, which may limit access to the vertex.

Status and justification. The sign table of the factors of $a(x-r)(x-s)$ gives the result. The vertex form compares $f(x)$ with β through the sign of $a(x-\alpha)^2$.

Example 4. $-2(x-1)(x-5) \geq 0$ is equivalent to $x \in [1,5]$. For $A(x) = x(12-x)$ on $[0,12]$, $A(x) = 36 - (x-6)^2 \leq 36$, with equality at 6. A rectangle with perimeter 24 m therefore has a maximum area of 36 m^2 when it is a square.

Watch out. In a strict inequality, the roots are excluded. Check the geometric bounds before concluding.

Application. Solve $x^2 - 4x - 5 < 0$, then maximise $x(8-x)$ on $[0,3]$.

Notes

Exercise path

Start with exercises 1 and 2. Then choose one of exercises 3 or 4 and do it without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher designates one. Exercise 8 is revisited after a delay.

Exercise 2.1 · Completing the square

Consider the function f defined on $\mathbb{R}$ by $f(x) = x^2 - 8x + 7$.

1. Complete the square to write $f(x)$ in the form $(x-4)^2 + k$, where k is a real number to be determined.
2. Deduce a factorisation of $f(x)$ using a special product.
3. Draw the sign table of f, then solve the inequality $f(x) \leq 0$ in $\mathbb{R}$.

Exercise 2.2 · Discussing without heavy calculation

We study the functions $f(x) = x^2 + 4$, $g(x) = (x+2)^2$ and $h(x) = -3(x-1)(x+4)$, all defined on $\mathbb{R}$.

1. For each function, determine any real roots by using the given form directly.
2. State the sign of each function on $\mathbb{R}$. State explicitly the values where it vanishes.
3. Explain why none of the three cases requires calculating the discriminant.

Exercise 2.3 · Choosing a strategy

We wish to solve the following three equations in $\mathbb{R}$:

$$x^2 - 9 = 0, \qquad x^2 - 9x + 20 = 0, \qquad 2x^2 - 3x - 1 = 0.$$

1. For each equation, choose a method: special product, finding two numbers with a known sum and product, or discriminant. Explain this choice.
2. Solve the three equations and give the solution sets in exact form.
3. Check the solutions of the first two equations by substitution. For the third, check the sum and the product of the roots obtained.

Exercise 2.4 · Recovering a parabola

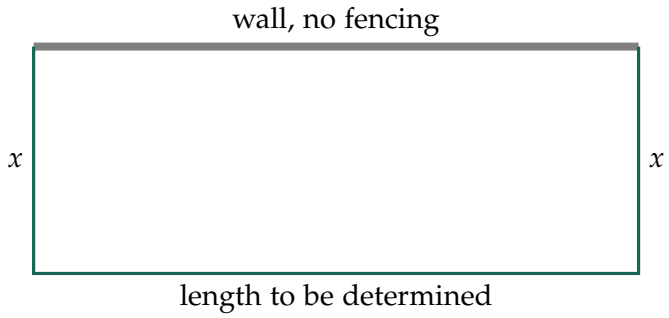
The curve of a polynomial function f of degree 2 crosses the x-axis at the points with x-coordinates -2 and 3. It also passes through the point $A(1, 12)$.

1. Determine an expression for $f(x)$ from these three pieces of information. State the form chosen and why it is suitable.
2. Expand the expression found. Deduce the y-intercept of the curve.
3. Check that the function obtained satisfies the three initial pieces of information.

Exercise 2.5 · A vegetable garden against a wall

A gardener has 20 m of fencing to enclose three sides of a rectangular vegetable garden. The fourth side runs along a wall and needs no fencing. Let x be the length, in metres, of each of the two sides perpendicular to the wall. All the fencing is used.

1. Make a sketch and express the length of the side parallel to the wall in terms of x. State the values of x for which the rectangle is not degenerate.
2. Express the area $A(x)$, in square metres. Determine its maximum value and the corresponding dimensions, using the method of your choice.
3. Check that these dimensions use exactly 20 m of fencing.



Sketch of the vegetable garden: only the three green lines use the 20 m of fencing. The proportions of the drawing do not fix the optimal dimensions.

Exercise 2.6 · A parameter

For each real number m, consider the equation in the real unknown x:

$$x^2 - 2x + m = 0.$$

The number m is a parameter: it stays fixed while the equation is solved.

1. Determine the values of m for which the equation has two distinct real solutions.
2. Determine the values of m for which it has a single real solution, then those for which it has none.
3. Give the solution in the case where it is unique. Explain why the three cases considered cover all values of m.

Exercise 2.7 · Link with sequences

A sequence is defined by $u_n = n^2 - 5n + 4$ for every natural number n. We look for all the indices at which the term of the sequence is strictly negative.

1. First solve the inequality $x^2 - 5x + 4 < 0$ in $\mathbb{R}$.
2. Deduce all the natural numbers n such that $u_n < 0$. Justify why a real interval is not the final answer here.

Exercise 2.8 · Delayed review

A student studies the function f defined on $\mathbb{R}$ by $f(x) = 2(x+3)^2 - 5$. She writes: "The minimum is -5 and it is reached at $x = 3$."

1. State what is correct and what is wrong in this sentence. Rewrite it correctly, justifying your answer.
2. Solve the inequality $f(x) \leq -3$ in $\mathbb{R}$, then check the two endpoints of the set obtained.

Homework. Do only the task designated at the end of the session, in the dated progression: 10–20 min at most. The exercises depend on the topics already studied, so some questions are deferred. Revisit exercise 8 one week after the chapter (10 min). If the allotted time is exceeded, note the difficulty and bring your attempts; do not pile up the whole reserve.

To do in the CQFD textbook

These exercises are done in your textbook. The progression indicates the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 43 p. 99, questions a and c. First find an obvious root, then the other using the product of the roots.

CQFD, exercise 49 p. 99, questions a and b. Translate the sum and the product into an equation; do not assume in advance that the numbers exist.

MCQ 1: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. The roots of $(x+1)(x-4)$ are
   - A. 1 and 4
   - B. -1 and -4
   - C. -1 and 4
   - D. 1 and -4
2. The discriminant of $x^2 - 2x + 3$ equals
   - A. -8
   - B. 8
   - C. 16
   - D. -2
3. The vertex of $(x-2)^2 + 5$ is
   - A. $(2, -5)$
   - B. $(-2, 5)$
   - C. $(5, 2)$
   - D. $(2, 5)$
4. The sum of the roots of $2x^2 - 6x + 1$ equals
   - A. 6
   - B. 3
   - C. -3
   - D. $\frac{1}{2}$
5. $x^2 - 4 < 0$ is equivalent to
   - A. $x > 2$
   - B. $x < -2$
   - C. $x \in (-2, 2)$
   - D. $x \in [-2, 2]$

MCQ 2: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $x^2 + 6x + 9$ has as its factorisation
   - A. $(x-3)^2$
   - B. $(x+9)(x+1)$
   - C. $(x+6)^2$
   - D. $(x+3)^2$
2. The product of the roots of $3x^2 + x - 2$ equals
   - A. $-\frac{2}{3}$
   - B. $-\frac{1}{3}$
   - C. $\frac{2}{3}$
   - D. -2
3. $-x^2 + 4x$ reaches its maximum on $[0,1]$ at
   - A. 2
   - B. 0
   - C. 1
   - D. 4
4. A double root requires
   - A. $a = 0$
   - B. $\Delta = 0$
   - C. $b = 0$
   - D. $c = 0$
5. The polynomials of degree 2 with roots 0 and 2 are
   - A. $a(x+2)$
   - B. $a(x-2)^2$
   - C. $x(x-2)$ only
   - D. $ax(x-2)$, $a \neq 0$

True or false? Justify.

1. A quadratic function always has two real roots.
2. A quadratic with two distinct roots is negative between them.
3. For every real number x, $(x-2)^2 \geq 0$.

> **Key point · Self-assessment.** I justify the choice of form; I distinguish roots from the vertex; I express the solution of an inequality as a set; I respect the domain of the model.
>
> For each objective, indicate: achieved without help; achieved with help; to be reworked. Write one example and one mistake to avoid.

Solutions and teaching notes for chapter 2

Teacher use. The answers below concern the tasks not worked in the lesson. First have students state their method; give graded help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. $x^2 - 6x + 9$. *Action:* Rebuild the product of two binomials.
2. $(x-4)(x+4)$. *Action:* Use the difference of two squares on two examples.
3. $x = 2$ or $x = -5$. *Action:* Have students say aloud the zero-product rule and its hypothesis.
4. $x = -3$ or $x = 3$. *Action:* Distinguish the positive square root from the solutions of the equation.
5. Negative before 3, zero at 3, positive after 3. *Action:* Draw the line, then make the sign table.
6. 4, reached only at 1. *Action:* Start from the fact that a square is non-negative.

Investigation activity

For width x, the length is $12 - x$; the geometric domain is $(0, 12)$. The requested areas are respectively 20, 32, 36, 32 and 20 square units. For every admissible width, $A(x) = x(12-x) = 36 - (x-6)^2 \leq 36$, with equality only at $x = 6$. Thus the maximum-area rectangle is a square of side 6. Trials suggest the result; the nonnegative square supplies the proof.

Immediate applications

1. $g(x) = (x+1)^2 - 4 = (x-1)(x+3)$. Roots 1 and -3, minimum -4 at -1.
2. The first is $3(x-1)^2 = 0$, so 1. For the second, $\Delta = -16 < 0$: no real solution.
3. $f(x) = a(x-1)(x-4)$; $4a = -8$, so $f(x) = -2(x-1)(x-4)$.
4. $(x-5)(x+1) < 0$ gives $(-1, 5)$. The vertex of $x(8-x)$ is at 4, outside the domain; on $[0, 3]$ the function is increasing and its maximum is 15 at 3.

Exercises: method and results

Exercise 2.1: Completing the square. $x^2 - 8x + 7 = (x-4)^2 - 9 = (x-1)(x-7)$. The leading coefficient is positive; the quadratic is negative or zero between the roots: $[1, 7]$.

Exercise 2.2: Discussing without heavy calculation. $x^2 + 4 > 0$ always, no root. $(x+2)^2 \geq 0$, double root -2. For the last one, roots -4 and 1; positive on $(-4, 1)$, negative outside, zero at the endpoints.

Exercise 2.3: Choosing a strategy. Special product: $x = \pm 3$. Sum and product: 4 and 5. For the third, no obvious integer root: $\Delta = 17$ and $x = \dfrac{3 \pm \sqrt{17}}{4}$. The discriminant works in every case, but is not always the most efficient.

Exercise 2.4: Recovering a parabola. $f(x) = a(x+2)(x-3)$. At 1: $-6a = 12$, so $a = -2$; $f(x) = -2x^2 + 2x + 12$, hence $f(0) = 12$.

Exercise 2.5: A vegetable garden against a wall. The two widths use $2x$ metres; length $20 - 2x$. Domain $[0, 10]$ for the function, $(0, 10)$ for a non-degenerate rectangle. $A(x) = 20x - 2x^2 = 50 - 2(x-5)^2$. Maximum 50 m^2, dimensions 5 m and 10 m.

Exercise 2.6: A parameter. $\Delta = 4 - 4m$. Two roots if $m < 1$; double root 1 if $m = 1$; none if $m > 1$. The parameter m stays fixed when we solve for x.

Exercise 2.7: Link with sequences. $u_n = (n-1)(n-4)$; negative for $1 < n < 4$. Since n is an integer, the only indices are 2 and 3. The set of indices is not the whole real interval.

Exercise 2.8: Delayed review. The minimum is -5 at -3. The inequality gives $(x+3)^2 \leq 1$, that is $-1 \leq x+3 \leq 1$, so $[-4, -2]$.

CQFD, exercise 43 p. 99, questions a and c. a: obvious root 1, product -30, so the second root is -30. c: write $4x^2 + 9x + 5$; root -1, product $\frac{5}{4}$, so the second root is $-\frac{5}{4}$. Check the sums, -29 and $-\frac{9}{4}$ respectively.

CQFD, exercise 49 p. 99, questions a and b. The numbers are the roots of $t^2 - st + p$. a: $t^2 - t + 1$, discriminant -3: no real pair. b: $t^2 + 60t + 756$, discriminant $576 = 24^2$: -18 and -42. Their sum is -60 and their product 756.

MCQ: answers and reasons

MCQ 1.

1. C. Set each factor to zero.
2. A. $4 - 12 = -8$.
3. D. The square vanishes at 2, value 5.
4. B. $-\frac{b}{a} = \frac{6}{2} = 3$.
5. C. Positive leading coefficient, strict inequality.

MCQ 2.

1. D. Special product.
2. A. $\frac{c}{a} = -\frac{2}{3}$.
3. C. The function increases up to 2; on this domain, right endpoint 1.
4. B. For a nonzero, a zero discriminant characterises the double root.
5. D. The multiplying coefficient is free and nonzero.

True or false

1. False: $x^2 + 1$ is strictly positive.
2. False without a hypothesis on a; $-(x-1)(x-2)$ is positive between 1 and 2.
3. True: the square of a real number is positive or zero; equality only at 2.

Differentiated teaching. For a technical block, offer only the first step (domain, factor or reference sample space), then withdraw the help. For an independent student, ask for a second method and the treatment of a limiting case. Orally, allow an initial reformulation in the language the student has mastered, then require a precise mathematical sentence in French.